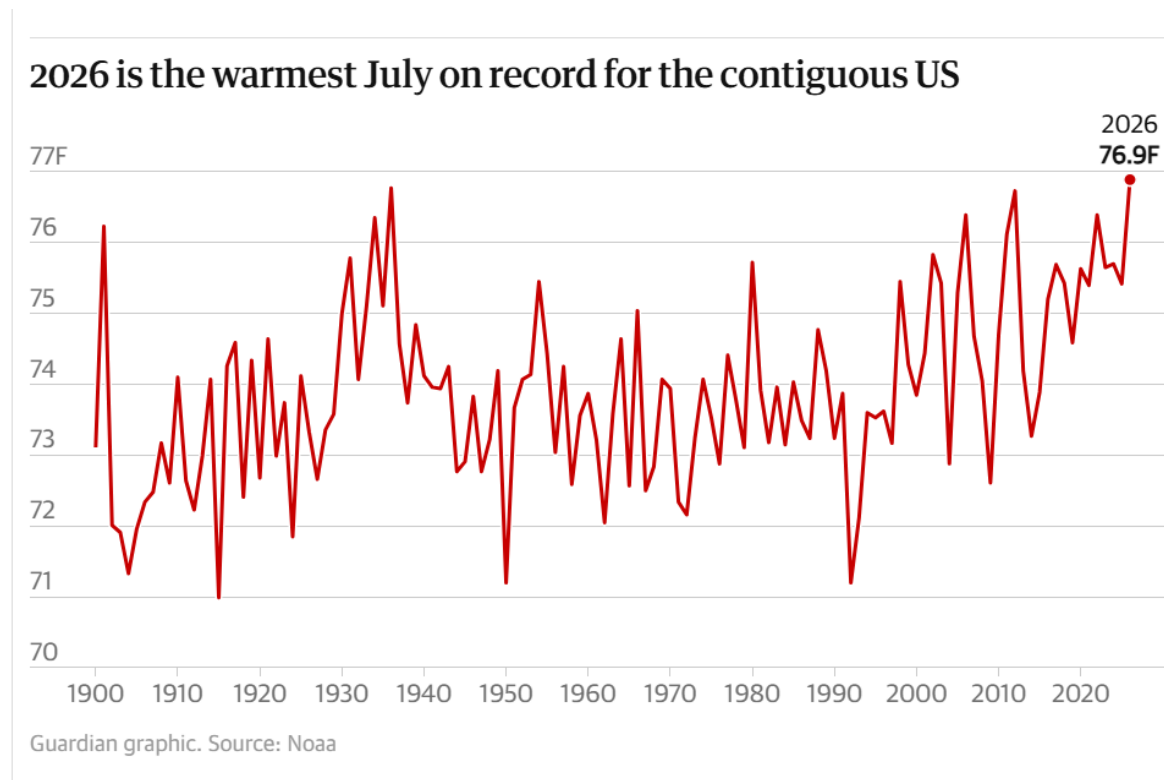


A critique on a chart in published work



Source: <https://www.theguardian.com/us-news/2026/aug/10/july-hottest-month-ever-recorded-heatwaves>

The graph, published in *The Guardian*, shows the record-breaking July temperatures and extreme heat recorded across the US. The x-axis shows the years, and the y-axis shows temperature; the line connecting the yearly values shows changes in temperature over time. However, some limitations can be identified in the way the data has been presented in the graph.

The x-axis shows the yearly variation in temperature from 1900 to 2026, which covers a long period, thus displaying a large number of year-to-year fluctuations in the graph. Although these variations accurately represent changes in July temperatures, the frequent ups and downs in the line make it difficult to identify a clear overall trend in temperature changes over time. As a result, the long-term pattern is not immediately visible to the reader. Adding a trend line or a 5- or 20-year moving average would make the overall long-term pattern easier to see.

The graph also shows that the difference between the 2026 record and previous record-high temperatures is very small. In July 2026, the temperature was 76.9°F, 0.1°F higher than 1936 (76.8°F) and 0.2°F higher than 2012 (76.7°F). Although 2026 looks like a new record, the difference from previous records is not very significant. The article states that the July 2026 temperature was 3.3°F above the 20th-century average. Showing this average directly on the graph as a reference line would provide a clear baseline for comparison and show how unusual the 2026 temperature was, communicating the magnitude of the increase more clearly.